

Research Proposal Submitted to Q-Group

Mutual Fund Trading Costs

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Abstract

The profitability of an investment strategy depends on both the information value and the implementation costs of the strategy. While the crucial importance of trading costs is well-recognized in practice, relatively little has been known about the variation in trading costs across portfolio managers and its impacts on portfolio performance. One of the difficulties in understanding these relations is the lack of data that allow joint analysis of portfolio characteristics/performance and trading costs. In this study, we overcome this data limitation by linking together three commonly-used databases: (1) The ANcerno database of institutional trades, which contains detailed trade records of a large number of institutional investors; (2) Thomson mutual fund holdings database, which contains quarterly holdings of US stocks by mutual funds; (3) CRSP survivor-bias-free mutual fund database, which reports monthly return, total net asset value, and other fund level information such as investment objective, expense ratio and turnover.

Our matched sample includes more than 500 US equity funds from 1999 to 2010. This large panel data set allows us to measure the trading costs directly at the trade level, and link them to the fund characteristics and performance. Using this data set, we investigate two closely-related questions:(1) How do trading costs differ across funds? (2) How do trading costs affect fund performance?

To the best of our knowledge, we are the first to examine the cross-section of equity trading costs among US mutual funds using trade-by-trade data. Previous studies on institutional trading costs are either at the institution level, with limited information about the institutions, or rely on trades inferred from quarterly holdings data and ignore heterogeneity in trading skills across funds. Our trading costs estimated directly from the actual trade prices draw a more accurate picture of the “invisible costs” of mutual fund trading.

Our results are highly relevant for mutual fund managers, investors, and researchers. Specifically, the cross-sectional relation we obtain between trading costs and various fund characteristics provides a simple tool to estimate mutual fund trading costs and evaluate their trading performance. A researcher who has no access to the actual trade data can estimate the trading costs of a mutual fund by applying our regression results. Similarly, a fund management company can detect whether a fund’s trading costs are “abnormally” high or low using our regression results as a benchmark. Furthermore, our results on the relation between trading costs and fund performance can be a valuable guide toward mutual fund investment.

Our results also provide important insights into the nature of active portfolio management. For example, the relation between fund/family size and trading costs tells us whether economy or diseconomy of scale dominates in institutional trading. The relation between fund performance and trading costs allows us to quantify the contribution of trade execution to fund performance and to assess whether the information value of mutual fund strategies tend to be offset by their implementation costs.

1 Research Questions

The profitability of an investment strategy depends on both the information value and the implementation costs of the strategy. While the crucial importance of trading costs is well-recognized in practice, relatively little has been known about the variation in trading costs across portfolio managers and its impacts on portfolio performance. One of the difficulties in understanding these relations is the lack of data that allow joint analysis of portfolio characteristics/performance and trading costs. In this study, we overcome this data limitation by linking together three commonly-used databases: (1) The ANcerno database of institutional trades, which contains detailed trade records of a large number of institutional investors; (2) Thomson mutual fund holdings database, which contains quarterly holdings of US stocks by mutual funds; (3) CRSP survivor-bias-free mutual fund database, which reports monthly return, total net asset value, and other fund level information such as investment objective, expense ratio and turnover.

Our matched sample includes more than 500 US equity funds from 1999 to 2010. This large panel data set allows us to measure the trading costs directly at the trade level, and link them to the fund characteristics and performance. Using this data set, we investigate two closely-related questions:

1. How do trading costs differ across funds?
2. How do trading costs affect fund performance?

By examining the cross-sectional variations of trading costs and their impact on fund performance, we can gain important insights into a number of issues highly relevant for active portfolio management. For example, does institutional trading exhibit decreasing return to scale, as postulated by Berk and Green (2004), in the sense that bigger funds face higher trading costs? Do funds trading more actively or charging higher fees trade at higher or lower costs? Are funds better at trade execution also better at stock selection? How persistent are the cross-sectional differences in trading costs? What fraction of fund performance can be explained by trading costs? Answers to these questions can lead to a better understanding of the nature of professional portfolio management, and provide valuable guidance toward mutual fund investment.

2 Methodology

Our study begins with a tedious process of merging three databases. The ANcerno institutional trading database contains detailed trading records of 1453 institutional clients, mainly pension fund sponsors and mutual fund managers, over the period 1999 to 2010.¹ Each observation in the database represents an execution, with information about the traded stock, the execution price, the number of shares traded, and whether the trade is a buy or sell, etc. Each institution is identified by a unique client code, while each trading unit within an institution, such a mutual fund in a fund family, is identified by a client-mgr code, which may change over time. The names of the institutions and funds are not disclosed.

We match the ANcerno clientmgr codes to the mutual funds in the Thomson holdings database by exploring the relation between stock (holdings) and flow (trades). In an ideal situation, changes in stock holdings over a reporting period should reflect all trades during the period. Based on this idea, we adopt the following matching procedure. First, for each Thomson mutual fund, we identify all changes in stock holdings during a reporting period (mostly at the quarterly interval), adjusted for stock splits and stock dividends. Second, for each ANcerno clientmgr code, we aggregate all its trades during a Thompson reporting period, and calculate the implied changes in stock holdings over that period. Third, we apply a series of criteria to identify an ANcerno-Thompson match by matching the holding- and trade-implied portfolio changes. The matched ANcerno-Thompson funds are then matched to the CRSP mutual fund database using the MFlink file originally developed by professor Russ Wermers and available through WRDS.

Following Anand, Irvine, Puckett, and Venkataraman (2011), we measure the trading costs by execution shortfall and calculate it on a daily basis, stock by stock and fund by fund. More specifically, the trading costs for a given stock in a given trade direction (buy or sell) on date t is defined as

$$TC(t) = \frac{P_1(t) - P_0(t)}{P_0(t)} * D(t) \quad (1)$$

where $P_1(t)$ measures the value-weighted execution price of on date t , $P_0(t)$ is the price

¹See Anand, Irvine, Puckett, and Venkataraman (2012) for a brief description of this database.

at the open of the day, and $D(t)$ is a variable that equals 1 for a buy and equals -1 for a sell. As a robustness check, we also use the percentage deviation from the value-weighted average price (VWAP) as an alternative measure of trading costs.

We value-weight the execution shortfall across days and stocks to get the average trading cost for each fund at the quarterly interval. We then multiply the average trading cost by the turnover rate reported in the CRSP mutual fund database to get a fund's quarterly trading costs in percentage.

After computing the measure of trading costs for each fund, we regress it on a number of fund characteristics to investigate the cross-sectional determinants of trading costs. These characteristics include: (1) fund investment style (value vs. growth, large cap vs. small cap etc); (2) fund size; (3) turnover; (4) expense ratio; (5) fund inflow; (6) fund family size. These regressions provide insights about which funds are better at minimizing trading costs. For example, they tell us whether bigger funds or fund families suffer from diseconomies of scale by facing higher trading costs, or whether they are able to trade cheaply because of their access to more sophisticated trading techniques and richer resources. They also tell us whether funds charging higher explicit fees pay higher or lower "invisible costs" in trading, and how fund flow volatility affects trading costs. Furthermore, given that turnover is often used as a proxy for trading costs, it is interesting to know how strongly turnover is related to the transaction costs we measure directly using the trade data.

We also examine the persistence of trading costs using autoregressive regressions and transition tables. While it is well-known that good performance of mutual funds tends to be short-lived, it is an interesting question whether some funds are able to consistently trade at a lower cost.

We then examine the relation between fund performance and trading costs. We measure fund performance by the standard metrics such as the three- or four-factor alphas, and regress them on trading costs. We control for other fund characteristics potentially affecting fund performance, such as fund size and age. A negative relation between performance and trading costs suggests that the information value of high-cost strategies is not sufficient to cover their implementation costs, while an insignificant relation would support the notion that the information value of mutual fund strategies tend to be offset by their implementation costs.

3 Related Literature

Our paper merges two lines of research. The first strand of literature examines trading costs at the institution level using trade data. One pioneering study is by Keim and Madhavan (1997), who examine the equity transactions of 21 institutions from 1991 to 1993. They find that trading costs are economically significant, and vary with both trader-specific factors such as investment style and order submission strategy, as well as stock-specific factors such as exchange listing. Two recent papers along this line are Anand, Irvine, Puckett, and Venkataraman (2011) and Anand, Irvine, Puckett, and Venkataraman (2012). They use the ANcerno trade data as we do, but perform analysis at the institution rather than fund level. Both studies find that some institutions can trade at a lower cost consistently over time. The latter also finds that an institution's trade execution skills are positively related to its stock selection abilities, measured by the future performance of traded stocks. While these studies provide valuable insights into costs of institutional trades, they share a common limitation: they have very limited information about the characteristics of the trading institutions. As a result, they are not very helpful in understanding the cross-sectional determinant of trading costs. Furthermore, given that the analysis is done at the level of institutions, which typically manage a large number of portfolios, they do not link the trading costs to the performance of the underlying portfolios. Therefore, they are silent about how trading costs affect fund performance. Our data set allows us to fill both gaps of these studies.

The second strand of literature focuses on the impact of trading costs on mutual fund performance. With the notable exception of Christoffersen, Keim, and Musto (2006), none of the studies that we are aware of measures trading costs directly using trade data.² For example, Carhart (1997) uses portfolio turnover as a proxy for trading costs and finds it negatively related to fund performance. However, Edelen, Evans, and Kadlec (2011) find that turnover is a poor proxy for trading costs. A number of studies employ a trade-by-trade aggregation to estimate fund trading costs by combining stock-specific cost estimates with trading activities inferred from changes in quarterly portfolio holdings. For example, Chalmers, Edelen, and Kadlec (1999) takes this approach and use a fund's

²See Edelen, Evans, and Kadlec (2011) for a review of this literature.

brokerage commissions and stock's effective bid-ask spread as a proxy for trading costs. Their measure ignores price impact, an important component of trading costs. Wermers (2000) and Kacperczyk, Sialm, and Zheng (2008) estimate mutual fund trading costs using the regression coefficients estimated by Keim and Madhavan (1997). Edelen, Evans, and Kadlec (2011) use a measure that combines changes in quarterly holding data with trading costs estimated for each stock from NYSE TAQ data. A common problem of these holding-based approaches is that changes in quarterly holdings do not capture interim trades within a quarter. More importantly, these studies implicitly assume that the same trading costs apply to all funds trading the same stock, and totally ignore the heterogeneity in trading skills across funds. Our trading costs measured directly at the trade level are free of both these problems.

Christoffersen, Keim, and Musto (2006) analyze the trade data of 210 Canadian mutual funds from 2001 to 2003. They find that active management and bigger fund size are associated with both cheaper trades and better subsequent performance (measured by the performance of traded stocks). Compared to our study, their sample is relatively small, and they do not examine the relation between trade performance and the actual performance of the fund.

4 Contribution

Our study contributes to the literature in several dimensions:

First, to the best of our knowledge, we are the first to examine the cross-section of equity trading costs among US mutual funds using trade-by-trade data. Previous studies on institutional trading costs are either at the institution level, with limited information about the institutions, or rely on trades inferred from quarterly holdings data and ignore heterogeneity in trading skills across funds. Our trading costs estimated directly from the actual trade prices draw a more accurate picture of the “invisible costs” of mutual fund trading.

Second, our results are highly relevant for mutual fund managers, investors, and researchers. Specifically, the cross-sectional relation we obtain between trading costs and various fund characteristics provides a simple tool to estimate mutual fund trading costs

and evaluate their trading performance. A researcher who has no access to the actual trade data can estimate the trading costs of a mutual fund by applying our regression results. Similarly, a fund management company can detect whether a fund's trading costs are "abnormally" high or low using our regression results as a benchmark. Furthermore, our results on the relation between trading costs and fund performance can be a valuable guide toward mutual fund investment.

Third, our results provide important insights into the nature of active portfolio management. For example, the relation between fund/family size and trading costs tells us whether economy or diseconomy of scale dominates in institutional trading. The relation between fund performance and trading costs allows us to quantify the contribution of trade execution to fund performance and to assess whether the information value of mutual fund strategies tend to be offset by their implementation costs.

5 Timetable

Our work is ongoing at the time of this submission. We expect to have the first set of results by December 2012. Further robust checks should be done and a complete manuscript should be available by June 2013.

6 CVs of Researchers (attached)

References

- Anand, Amber, Paul Irvine, Andy Puckett, and Kumar Venkataraman, 2011, Market Crashes and Institutional Trading, Working paper, Syracuse University.
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Education:

1973-1977	Cornell University B.S. - Operations Research (with distinction)
1977-1978	Cornell University Johnson Graduate School of Management M.B.A. - Finance (with distinction)
1985-1990	Cornell University Johnson Graduate School of Management Ph.D. Finance Ph.D. Thesis: Credit Market Incentives and Constraints Thesis committee: Maureen O'Hara (chair), Robert Jarrow, and David Easley

Academic Positions:

1985-1990	Research Assistant, Department of Finance, Johnson Graduate School of Management, Cornell University
1990-1997	Assistant Professor, Department of Finance, Investment and Banking, School of Business, University of Wisconsin-Madison
1997-2006	Associate Professor, Department of Finance, Investment and Banking, School of Business, University of Wisconsin-Madison
2006-	Professor, Department of Finance, Investment and Banking, School of Business, University of Wisconsin-Madison
1998-2007	Aschenbrener Faculty Scholar, Puelicher Center for Banking Research
2007-	Jeffrey J. Diermeier Chair of Finance and Academic Director of the Hawk Center for Applied Security Analysis

Professional Experience:

1978-85	<u>Financial Analyst / Supervisor</u> <i>Ford Motor Co., Dearborn, Mi.</i> Analyzed alternative product and manufacturing plans using capital budgeting techniques. Reviewed business plans and budget performance of various operations.
1987-88	<u>Chief Financial Officer</u> <i>V.I.P. Structures Inc., Syracuse N.Y.</i> Developed limited partnership for private placement. Converted to subchapter S status. Developed financial planning and budgeting systems integrated with monthly forecasts and actual results.
2000-01	<u>Chief Economist, Director of the Office of Economic Analysis</u> <i>U.S. Securities and Exchange Commission</i> Economic analysis in support of Commission rulemaking.

Journal Affiliations:

Associate Editor, *Review of Financial Studies* July 2001-July 2004

Associate Editor, *Financial Management* September 2002-October 2011

Honors and Awards:

1986, AACSB/GMAC National Doctoral Fellowship awarded semiannually to the outstanding Cornell Business School Ph.D. student

1988, Martin M. McVoy, Jr. Trust Fellowship awarded annually in a university-wide competition to the outstanding Ph.D. student who had achieved dissertator status

1994, "Volume, Volatility and NYSE Trading Halts" Nominated for the Smith Breeden Prize for the best paper of the year in the *Journal of Finance*

1997, "On the Robustness of Size and Book-to-Market in Cross-Sectional Regressions" Nominated for the Smith Breeden Prize for the best paper of the year in the *Journal of Finance*

Publications:

Lee, C.M.C. and M.J. Ready, 1991, "Inferring Trade Direction from Intraday Data," *Journal of Finance* 46, pp. 733-746.

Summary: This paper documents two potential problems with classifying trades based on the current bid and ask quotes: new quotes may be recorded ahead of trades that triggered them and trades inside the spread are not readily classifiable. These problems are analyzed in the context of the interaction between agents on the floor of the exchange. We show that comparing each trade to the quote in effect 5 seconds before the trade can eliminate most of the timing problems, and that the tick test can be useful in classifying trades at the midpoint of the spread.

Lee, C.M.C., B. Mucklow and M.J. Ready, 1993 "Spreads, Depths and the Impact of Earnings Information: An Intraday Analysis," *Review of Financial Studies* 6, pp. 345-374.

Summary: This paper investigates the changes in liquidity around earnings announcements. We hypothesize that there is a greater risk of informed trading during these periods, and that specialists should widen their spreads in response to this risk. We also suggest that specialists can reduce their risk by reducing the depth (or size) of the quotes, and ours is the first paper to empirically estimate changes in depth in response to increases in adverse selection costs. We show that spreads widen and depths fall in advance of earnings announcements, indicating that specialists actively manage both dimensions of the quote to control their adverse selection costs.

Publications (continued):

Lee, C.M.C., M.J. Ready and P.J. Seguin, 1994, "Volume, Volatility and NYSE Trading Halts," *Journal of Finance* 49, pp. 183-214.

This paper was translated into Japanese by Professor Keiichi Omura and published in *Investment* (a journal published by the Osaka Securities Exchange), 1994, pp. 2-36. An abstract of this paper was prepared by Professor Jan R. Squires, CFA, and published in *The CFA Digest*, August 1994, pp. 43-45.

Summary: We analyze the effectiveness of trading halts as a mechanism for the market to absorb information with the least possible disruption. We compare trading halts with "pseudo-halts", which are non-halt control periods matched on time-of-day, duration and absolute net-of-market returns. We show that volume (volatility) in the first full trading day *after* a trading halt is 230% (50-115%) higher than following the matching "pseudo-halts". These results are robust over different halt types and news categories. We also show that media coverage increases the volume and volatility following both halts and pseudo-halts, but a separate halt effect remains after controlling for the media effect. In summary, it appears that the halt mechanism is inferior to continuous trading as a way to allow the markets to digest new information.

Ready, M.J., and R.C. Ready, 1995, "Optimal Pricing of a Depletable Replaceable Asset: The Case of Landfill Tipping Fees," *Journal of Environmental Economics and Management* 28, pp. 307-323.

Summary: We develop a general model for pricing a depletable, replaceable asset, and discuss its application in determining optimal tipping fees for a sanitary landfill. We show that the optimal fee includes a component that grows at the real interest rate as space in the landfill is used, and then drops when a new landfill is built. Because the marginal cost of landfill space increases as the landfill is filled, it may be optimal to delay a waste reduction program (such as recycling, composting, or incineration) until the landfill is nearly full.

Knez, P.J. and M.J. Ready, 1996, "Estimating the Profits from Trading Strategies," *Review of Financial Studies* 9, pp. 1121-1164.

An abstract of this paper was published in *Journal of Finance* 51, July 1996, pp. 1055-1056.

Summary: We show that expected price improvement (the expected difference between the trade price and the posted bid or ask quote) is a non-linear function of both the size of the order and the quoted depth. This dependence is important for the estimation of profits from trading strategies because it means that the profits will depend critically on the size of the positions taken. Using the history of quotes and our estimates of expected price improvement, we simulate the profitability of a trading strategy that switches between small and large firm portfolios in an attempt to exploit the predictability of small firm returns. We find that the strategy would be swamped by the transaction costs, and we conclude that the predictability persists because it can't be profitably exploited.

Publications (continued):

Knez, P.J. and M.J. Ready, 1997, "On the Robustness of Size and Book-to-Market in Cross-Sectional Regressions," *Journal of Finance* 52, pp. 1355-1382.

An abstract of this paper was published in *Contemporary Finance Digest*, 1997, volume 1.

Summary: Fama and French (1992 and 1993) argue that size and book-to-market play a dominant role in explaining cross-sectional differences in expected returns. We extend the Fama and French (1992) cross-sectional regressions of returns on size and book-to-market to include a robust regression estimator that is not sensitive to outliers or leverage points. We find that the estimated risk premium on size is entirely driven by extreme observations representing less than 1% of the sample. We argue that investigating the few small firms that experience large positive returns could lead to an understanding of the economic forces underlying the size effect, and may also yield important insights into how firms grow.

Ready, M.J., 1999, "The Specialist's Discretion: Stopped Orders and Price Improvement" *Review of Financial Studies*. 12, pp.1075-1112.

Summary: When a market order arrives, the NYSE specialist can offer a \$1/8 better price to gain priority over the limit orders on the book and trade for his own account. Alternatively, the specialist can "stop" the market order, which means he guarantees execution at the current quote but provides the possibility of price improvement. I develop a model to show that specialists can use stops to sample the future order flow before making a commitment to trade, and I find empirical support for the model in that stops are most frequent following trades at the midpoint of the spread. I also present evidence that "unadvertised" liquidity, in the form of both stops and immediate price improvement, imposes adverse selection costs on limit order traders.

Ready, M.J., 2002, "Profits from Technical Trading Rules" *Financial Management* 31, pp. 43-61.

Summary: I compare the genetic-algorithm-based technical trading rules developed by Allen and Karjalainen (1998) with moving average rules studied by Brock, Lakonishok, and LeBaron (1992). I find two separate results consistent with the hypothesis that the popularity of the moving average rules stemmed from their spurious success in sample. First, I show that average stock returns for 1987-1998 (Brock, Lakonishok, and LeBaron's sample ends in 1986) were higher on days when these strategies chose to be out of the market. Second, I construct genetic-algorithm-based technical trading rules that focus on what appear to be the most promising patterns in the 1957-1962 data, and show that these rules under perform the moving average rules for 1963-1986.

Publications (continued):

Hanna, J.D. and M.J. Ready, 2005, "Profitable Predictability in the Cross Section of Stock Returns" *Journal of Financial Economics* 78, pp. 463-505.

Summary: Haugen and Baker (1996) report that a long-short stock selection strategy based on more than 50 measures of accounting information and past return behavior would have generated excess returns of approximately 3 percent per month. We find that the Haugen and Baker strategies do not provide attractive returns after transaction costs if an investor already has access to strategy portfolios based on book-to-market and momentum. We also provide an extensive analysis of transaction costs over a long sample and we report results of independent interest to researchers in market microstructure.

Odders-White, E.R. and M.J. Ready, 2006, "Credit Ratings and Stock Liquidity" *Review of Financial Studies* 19, pp. 119-157.

Summary: We analyze contemporaneous and predictive relations between credit ratings and measures of equity market liquidity, and find that common measures of adverse selection, which reflect a portion of the uncertainty about future firm value, are larger when credit ratings are poorer. We also show that future ratings changes can be predicted using current levels of adverse selection. Collectively, our results validate widely used microstructure measures of adverse selection, and offer new insights into the value of credit ratings and the specific nature of the information they contain.

He, C., E.R. Odders-White, and M.J. Ready, 2006, "The Impact of Preferencing on Execution Quality" *Journal of Financial Markets* 9, pp. 246-273.

Summary: We examine the impact of preferencing on execution quality for NASDAQ and NYSE-listed stocks. Our theoretical model demonstrates that realized spreads are more reliable than effective spreads in the presence of preferencing, but even realized spreads are a poor measure of execution quality if the stocks being compared have different degrees of information asymmetry. We provide a new measure of the costs of preferencing that is independent of asymmetric information. Using data from the SEC 11Ac1-5 reports for marketable orders of up to 2000 shares, we find that both realized spreads and our preferencing measure are lower for NYSE-listed stocks.

Odders-White, E.R. and M.J. Ready, 2008, "The Probability and Magnitude of Information Events" *Journal of Financial Economics* 87, pp. 227-248.

Summary: Models of adverse selection risk generally assume that market makers offset expected losses to informed traders with expected gains from the uninformed. We recognize that the expected loss captures a combination of two effects: 1) the *probability* that some traders have private information, and 2) the likely *magnitude* of that information. We use a maximum-likelihood approach to separately estimate the probability and the magnitude of private information events for NYSE-listed stocks from 1993 through 2003. The results shed light on the price discovery process and have implications for many areas of finance.

Current Work in Progress:

- “Determinants of volume in dark pools”, April 2012 working paper, presented at the 2010 meetings of the American Finance Association
- “An analysis of illegal insider trading” (with Diane Del Guercio and Elizabeth Odders-White), November 2011 working paper
- “Do investors overpay for stocks with lottery-like payoffs? An examination of the returns on OTC stocks”, (with Bjorn Eraker), October 2011 working paper

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Education

- 2001 – 2004 Ph.D in Finance, with Distinction, University of Vienna
Advisors: Josef Zechner, Engelbert Dockner
1994 – 1996 Master in Economics, Peking University
1989 – 1994 Bachelor in Economics, Peking University

Employment

- 2008 – Assistant professor, University of Wisconsin-Madison
2005 – 2008 Junior faculty member, University of Vienna
2001 – 2004 Research and teaching assistant, University of Vienna
2000 – 2001 Visiting scholar, Vienna University of Economics and Business
1996 – 2001 Lecturer, School of Economics, Peking University

Research Interests

Mutual Funds, Hedge Funds, Real Options, Empirical Asset Pricing

Selected Publications

Intermediated Investment Management, with Neal Stoughton and Josef Zechner, *Journal of Finance*, 2011, vol 66, no. 3, 947-980.

Market Discipline and Internal Governance in the Mutual Fund Industry (with Thomas Dangel and Josef Zechner), *Review of Financial Studies*, 2008, volume 21, no. 5, 2307-2343.

Working Papers

Mutual Fund Families and Performance Evaluation, with David Brown, 2011.

The Dynamics of Hedge Fund Fees, with Prachi Deuskar, Jay Wang, and Quoc

Nguyen, 2011.

Corporate Investment Over Uncertain Business Cycles, with Thomas Dangl, 2011.

Portfolio Performance, Discount Dynamics, and the Turnover of Closed-End Fund Managers, 2009, with Russ Wermers and Josef Zechner.

Concave Incentives and Delegated Portfolio Choice, 2008, with Jun Liu.

Disentangling Liquidity and Size Effects in Stock Returns: Evidence from China, 2008, with Rong Cui.

Academic Awards

University of Wisconsin-Madison Graduate School Research Competition Award, 2011, 2008

China Finance Review International Conference Best Paper Award (first prize winner), 2010

University of Cologne Center for Financial Research Outstanding Paper Award, 2008

The China Finance Association (TCFA) Best Paper Award (first prize winner), 2007

SWX Best Paper Award of Swiss Society for Financial Market Research, 2005

Peking University Outstanding Research Paper Award (Humanities and Social Sciences Group, second prize winner), 2000

Citibank Distinguished Teaching Award, Peking University, 1999&1997

Conference Presentations

American Finance Association meetings, 2012 (Chicago), 2009 (San Francisco), 2008 (New Orleans), 2006 (Boston)

Western Finance Association meetings, 2011 (Santa Fe), 2007 (Big Sky), 2006 (Keystone)

European Finance Association meetings, 2007 (Ljubljana), 2006 (Zurich), 2004 (Maastricht)

Utah Winter Finance Conference, 2012 (snowbird)

Financial Intermediation Research Society meetings, 2012 (scheduled, 2 papers, Minneapolis), 2011 (Sydney), 2008 (Anchorage), 2006 (Shanghai)

Utah Winter Finance Conference, 2012 (Salt Lake City, Scheduled)

22nd Annual Conference on Financial Economics and Accounting, 2011 (Indiana University)

China International Conference in Finance, 2011 (Wuhan, 3 papers), 2007 (Chengdu, 2 papers), 2005 (Kunming)

China Finance Review International Conference, 2010 (Shanghai)
European Financial Management Association meetings, 2007 (Vienna, 2 papers)
CEPR/Gerzensee European Summer Symposium in Financial Markets (Gerzensee), 2007, 2005
EMG conference on Emerging Market Finance, 2008 (City University London)
Colloquium on Financial Market, 2008 (University of Cologne)
Conference on professional asset management, 2007 (RSM Erasmus University)
Swiss Society for Financial Market Research meetings, 2005 (Zurich)
German Finance Association meetings, 2004 (Tübingen)

Seminar Presentations

University of Illinois Chicago, 2011
University of New South Wales, 2011
Technology University of Sydney, 2011
University of Hong Kong, 2011, 2005
Cheung Kong Graduate School of Business, 2011
University of Illinois Urbana Champaign, 2010
University of Wisconsin-Madison, 2011, 2010 (statistics and finance departments), 2007
Imperial College London, 2007
University of Cologne, 2007
Norwegian School of Management, 2006
University of Vienna, 2006, 2005, 2004
Peking University, 2006, 2005
Vienna Institute of Strategic Capital Market Research, 2005

Conference Discussant

ASU Winter Finance Conference, 2012 (Scottsdale)
Western Finance Association meetings, 2011 (Santa Fe)
Financial Intermediation Research Society meetings, 2011 (Sydney)
China International Conference in Finance, 2011 (Wuhan), 2007 (Chengdu)
Econometric Society North American Winter Meetings, 2009 (San Francisco)
EMG conference on Emerging Market Finance, 2008 (London)

European Finance Association meetings, 2007 (Ljubljana, 2 papers)
European Financial Management Association meetings, 2007 (Vienna, 2 papers)
Gutmann Symposium on Asset Management (Vienna), 2007, 2006
European Winter Finance Summit, 2007 (Lech)
Swiss Society for Financial Market Research meetings, 2005 (Zurich)
Western Finance Association meetings, 2004 (Vancouver)

Refereeing Activity

Journal Referee

Journal of Finance, Review of Financial Studies, Journal of Financial Economics, Review of Finance, Journal of Financial Intermediation, Journal of Empirical Finance, Journal of Banking and Finance, Finance Research Letters, European Journal of Finance, German Economic Review, Applied Economics, Corporate Governance: An International Review

Conference Reviewer

SFS Finance Cavalcade 2012
German Economic Association meetings in 2010
European Finance Association meetings in 2009
European Winter Finance Summit in 2009 and 2007
Gutmann Symposium on Portfolio Management in 2007, 2006, 2004

Other Refereeing Activity

Member of The China Finance Association Best Paper Award Committee in 2010, 2009 and 2008
Ad Hoc reviewer for a textbook on Derivatives (by Robert Jarrow and Arka Chatterjea), W. W. Norton & Company, Inc.
Oversee Reviewer for Research Grant Council (RGC) of Hong Kong

Teaching Experiences

Advance Derivatives (Master, UW-Madison)
Derivatives (undergraduate, UW-Madison)
International Financial Management (undergraduate, University of Vienna)
Asset Management (master, University of Vienna),
Valuation (master, University of Vienna)

Finance Paper Reading (Ph.D, University of Vienna)
Money and Banking (undergraduate, Peking University)
International Corporate Finance (executive, Peking University)
Financial Economics (master, Peking University)

Ph.D Thesis Advising (As Committee Member)

Wei Zhang, Tor-Erik Bakke, Bin Zhao, Tiantian Gu

University Services

Seminar organizer of UW-Madison, Department of Finance, 09/2008-06/2009, 09/2010-06/2011

Seminar organizer of University of Vienna and Vienna Graduate School of Finance, 2007/09-2008/06

Other Publications

English Journals

Banking Industry in China: Problems and Prospects, *IFAS Forum*, 1-2/2001, 4-9, Interdisciplinary Research Institute for Asian Studies, Vienna.

The SOE's Debt Problem and the Debt-to-Equity Swap (with Xiao Zhao), *China and World Economy*, 2000, Vol. 8, No. 5, 18-23.

Chinese Journals

The Multi-Equilibrium Approach to Understanding Currency Crises, *Economic Science*, Sept/Oct 2000.

Debt-to-Equity Swap: A Theoretical and Policy Analysis Based on Corporate Governance (with Xiao Zhao), *Economic Research Journal*, February 2000. Partly reprinted in *Chinese Social Science Digest* 2000.

The Self-Reinforcing Mechanisms of Deflation and the Implications for China, *Journal of World Economy*, November 1999. Reprinted in *Research on China's Deflation during 1998-2000*, edited by CCER Macro-Economic Group, Peking University Press, 2000.

Modern Theory of Currency Crises and its Implications, *Finance and Trade Economics*, March 1999. Reprinted in *Adjusting Macro-economic Policies and Maintaining Market Orientation*, edited by CCER Macro-Economic Group, Peking University Press, 1999.

Divergence and Alignment of Objectives Between Owners and Managers,

Economist, May 1996.

Insider's Control in State-Owned Enterprises: Causes and Solutions, **Reform**, July 1995. Reprinted in *Corporate Governance Structure: Theory and Practice*, edited by Beijing Law Society Economic Law Division, Mechanic Industry Press, 2004.

A Comparison of the Consumption Functions of Rural and Urban Households in China, **Economic Science**, May/June 1995.

Textbooks

Money and Banking (with Gang Yi, in Chinese), Shanghai People's Publishing House 1999.

Book Translations (from English to Chinese)

The Palgrave Dictionary of Money and Finance (with Jian Hu et al), Economic Science Press 2001.

Microeconomic Theory by Mas-Colell, Whinston and Green (with Jie Ma et al), Economic Science Press 2001.

Security Analysis by Benjamin Graham and David Dodd (with Wei Qiu, Chunrong Li and Zheng Huang), Hainan Press 1999.

Theory of Industrial organization by Jean Tirole (with Jie Ma et al), China Remin University Press 1998.

Selected Papers by James Mirrlees (with Jie Ma et al), The Commercial Press 1997.

Yue Wu

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Madison, WI 53706

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Email: yuewu@bus.wisc.edu

Education

Ph.D. Finance, Wisconsin School of Business, UW-Madison, expected May 2016

M.S. Computational Finance at Statistics, Purdue University, May 2011

M.S. Forestry and Natural Resources, Purdue University, Jul. 2009

B.S. Environmental Science and Engineering, Tsinghua University, Jul. 2007

Research Experience

Graduate Research Assistant Sep. 2009–Dec. 2010

Employer: Office of Institutional Research Purdue University

Analyze 2010 National Research Council (NRC) ranking of PhD programs and the associated statistical methodology.

Established logistic models to explain the relationship between student success and independent variables (demography, test scores, etc).

Built a linear regression model to retrospectively estimate high school ranking of Purdue freshmen.

Graduate Research Assistant May 2009–Sep. 2009

Advisor: Prof. Jose Figueroa-Lopez Purdue University

Developed a maximum likelihood estimation method for Levy financial market via short-term approximations.

Pre-processed a huge amount of intra-day stock data (from Bloomberg and Wharton Research Data Services) to produce inputs suitable for quantitative modeling.

Performed Levy Model calibration and validation.

Graduate Research Assistant Aug. 2007–May 2009

Advisor: Prof. Guofan Shao Purdue University

Developed an optimization algorithm for improving classification accuracy for land-use maps with proved efficiency on National Land Cover Dataset.

Established a spatial model for determining the sites with climate suitability for vineyards across China. (The results were adopted by a wine producer.)

Teaching Experience

TA of *Derivative Securities* at UW-Madison Spring 2012

Taught two discussion sessions every other week for 116 students with evaluation 4.61/5.

TA of *Advanced Derivative Securities* at UW-Madison Spring 2012

Graded course projects for 7 students with evaluation 5/5.

TA of *Derivative Securities* at UW-Madison Fall 2011

Taught two discussion sessions every week for 100 students with evaluation 4.74/5.

TA of *Elementary Statistical Methods* at Purdue Fall 2009

Taught 60 students to use SPSS for data analysis.

TA of *Fundamental Remote Sensing* at Purdue
Gave lectures and guided labs for 17 students.

Fall 2008

Selected Course Projects in Mathematical Finance

Pricing of knock-in Barrier call options by important sampling with antithetic variates
Calibration of Hull-White model and implementation of LIBOR market model
Valuation of American put options by Crank-Nicolson method with successive over-relaxation
Evaluation of European options using Heston volatility model
Pricing of Asian type option using Monte Carlo simulation
Application of Rubinstein's implied binomial tree model to IBM stock and its derivative
Covariance estimation of stock returns with an application to portfolio selection
The Distribution of Realized Stock Return Volatility
Stripping zero rates using bootstrap to determine instantaneous forward rate curve

Selected Coursework

Theory of Finance	Econometrics
Microeconomics	Macroeconomics
Probability	Stochastic Process
Mathematics of Finance	Advanced Probability and Options w. Numerical Methods
Options and Futures	Financial Management
Game Theory	Financial Time Series
Computational Statistics	Advanced Statistical Methodology
Mathematical Statistics	Partial Differential Equations

Leadership

Class president in Tsinghua Univ.: lead the class to achieve three university-level awards
Donation organizer: raised a tuition of \$1,500 for 50 students from poor families

Honors and Awards

Fellowship from Finance Department, Wisconsin School of Business, 2011
Lynn Fellowship, Purdue Univ., 2007
Excellent Graduate Award, Tsinghua Univ., 2007
First-Class Fellowship, Tsinghua Univ., 2004-2006

Test Scores

GRE: 530/800/4.0, June 2006 TOEFL: 107, Nov. 2008

Language

English, Fluent Chinese, Native